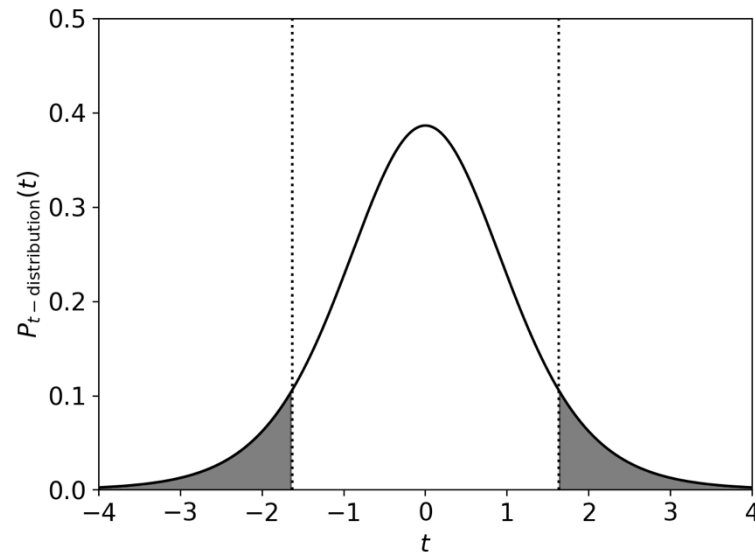
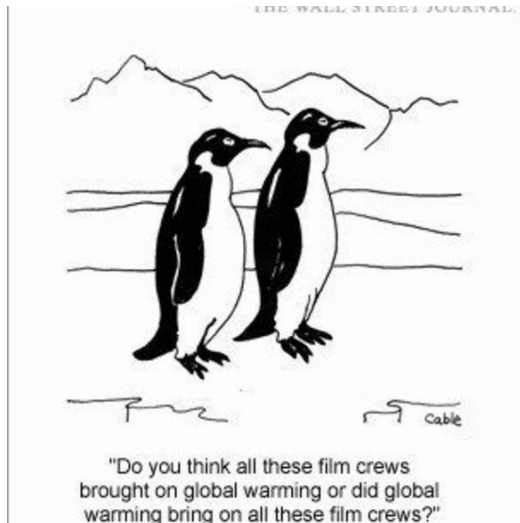
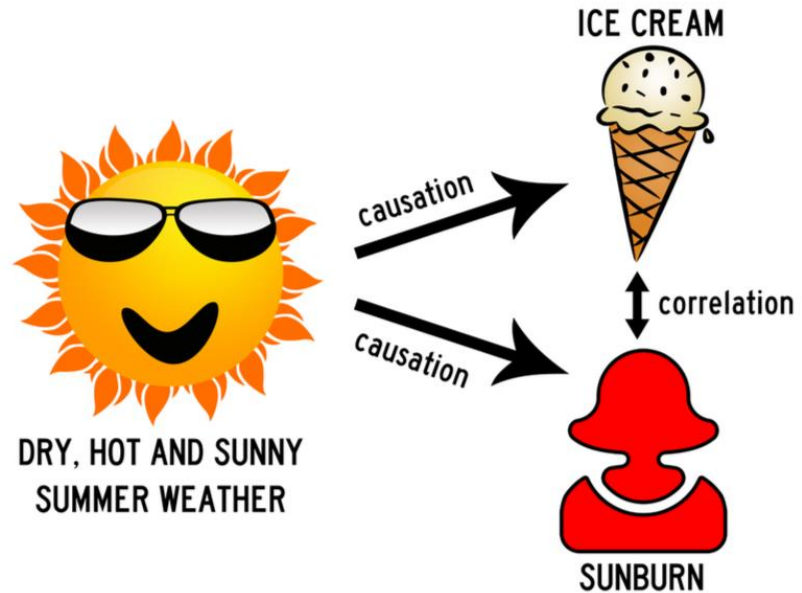
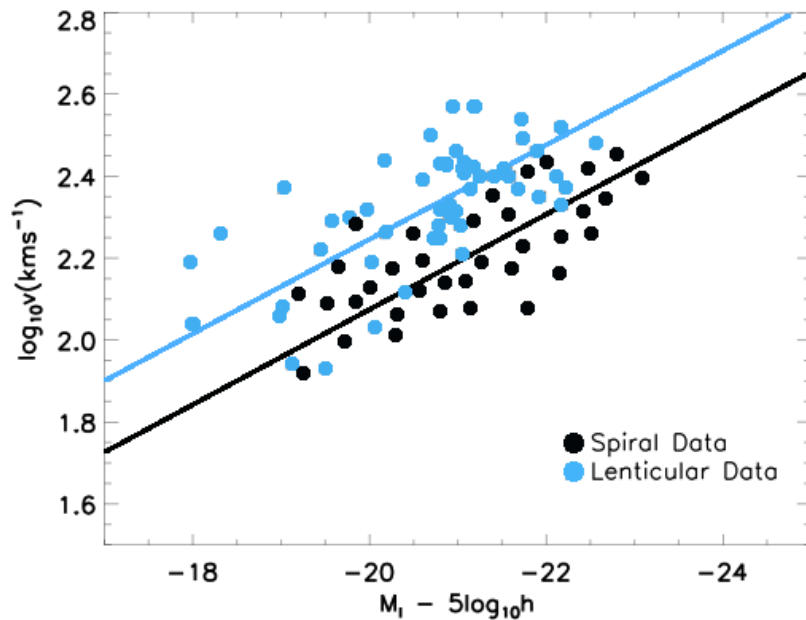
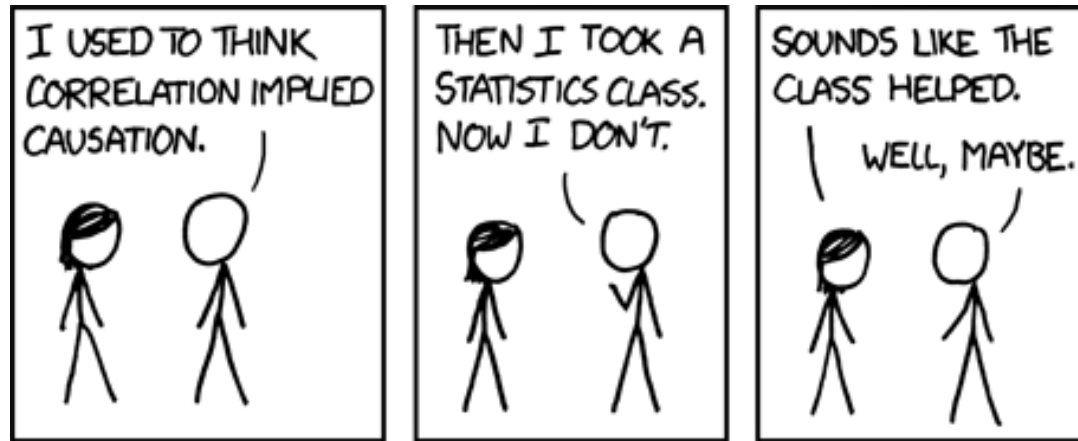


Class 2: Correlation Testing

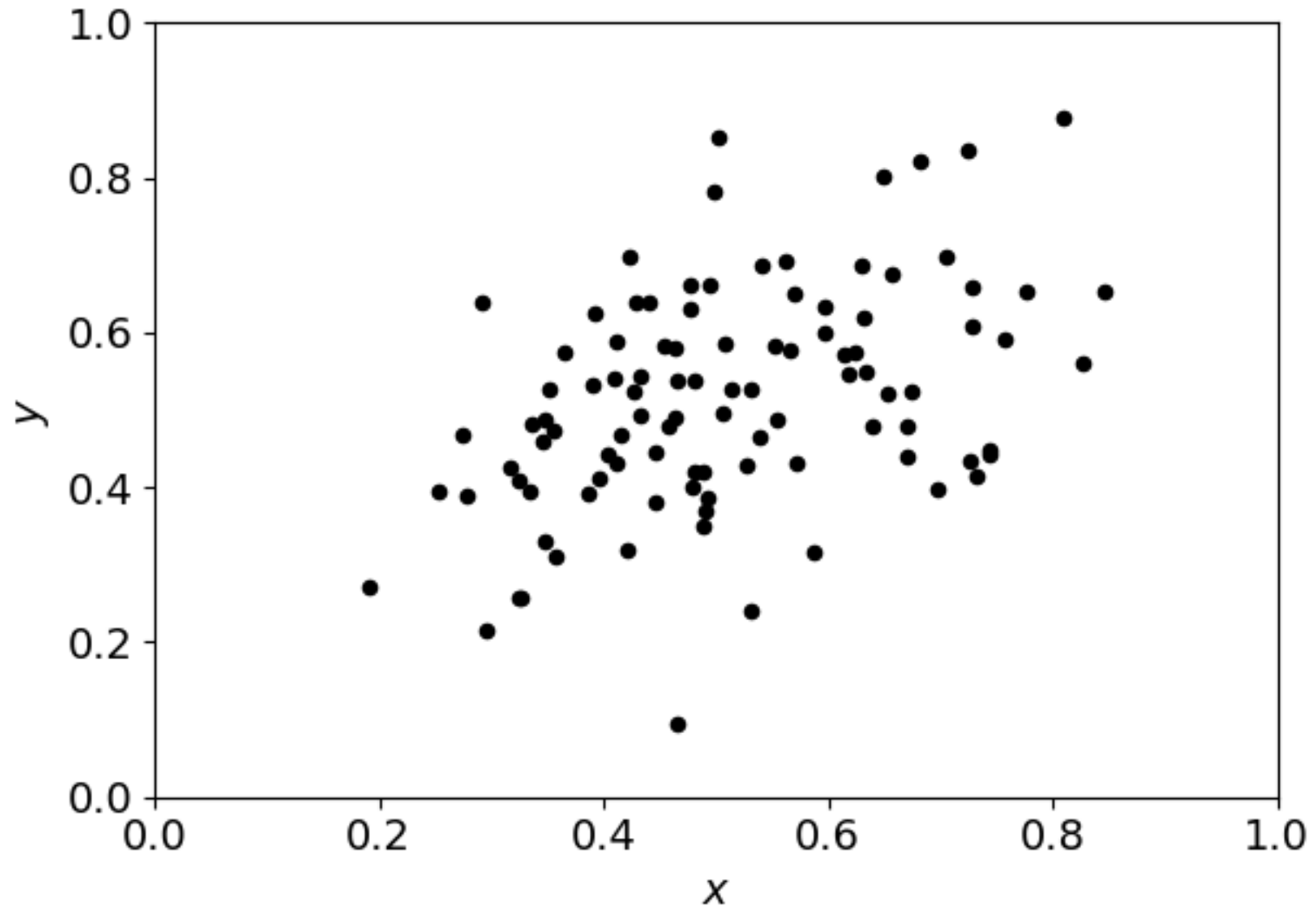


Quick intro to statistics course



- Week 1: Probability and statistics
- Week 2: Correlation testing
- Week 3: Model fitting
- Week 4: Regression
- Week 5: Error estimates
- Week 6: Bayesian methods

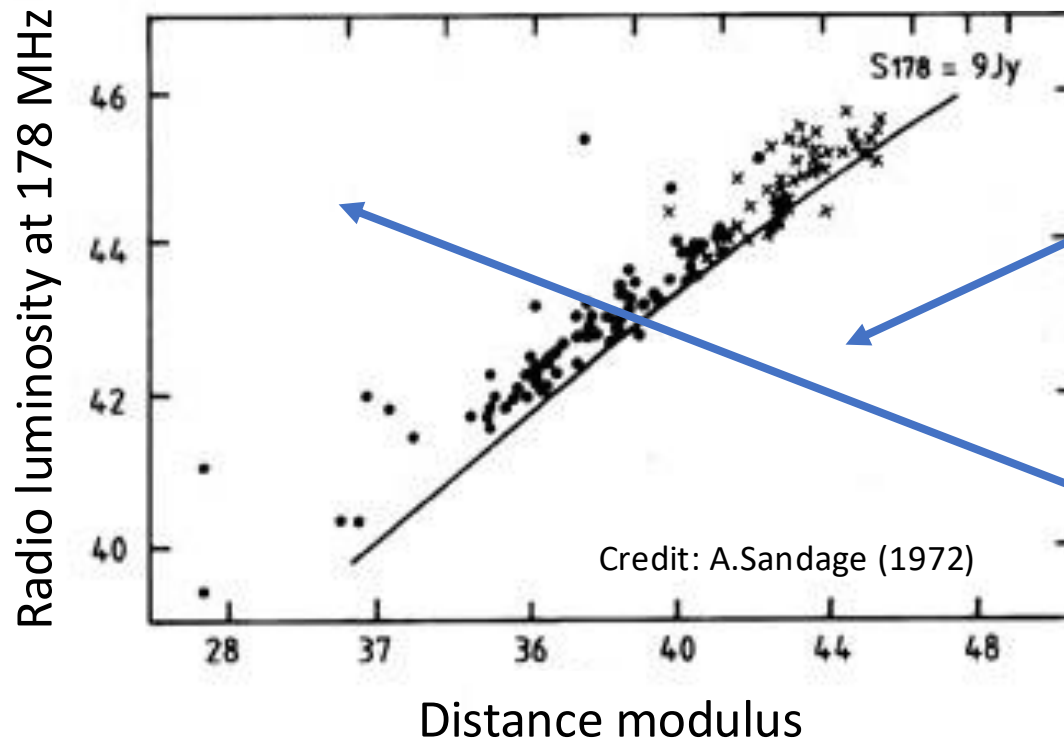
Searching for correlations!



Pitfalls when searching for correlations

Selection effects can easily lead to spurious correlations

Here is a perfect luminosity-redshift correlation for radio galaxies in the 3CR survey:



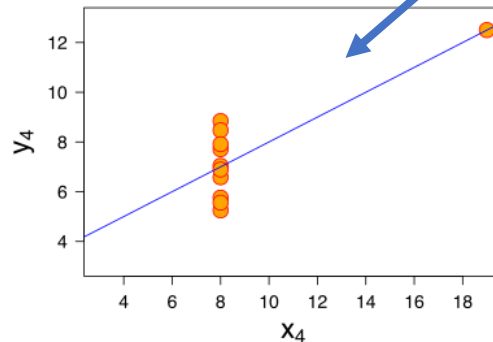
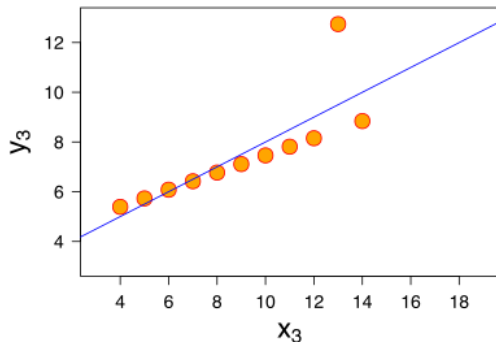
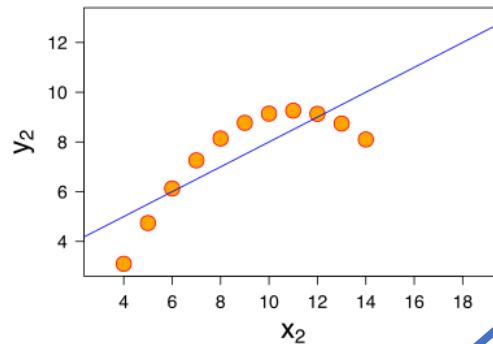
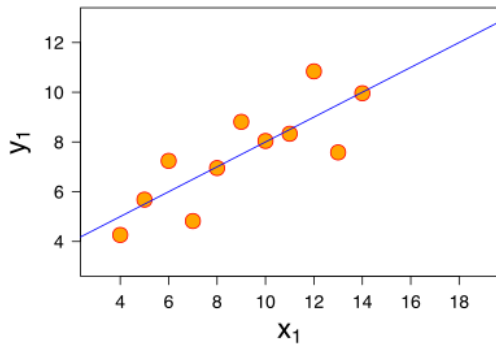
Galaxies cannot appear in this region because they are below the flux limit

Galaxies cannot appear in this region because of the steepness of the luminosity function

Pitfalls when searching for correlations

Correlations can be driven by a **small number of outliers**

The following four (x, y) datasets all have the same mean, variance, correlation coefficient and regression line:



The correlation here is completely driven by a single outlier

“Rule of thumb”: if a correlation goes away after you cover part of the dataset with your thumb, it probably isn’t real!

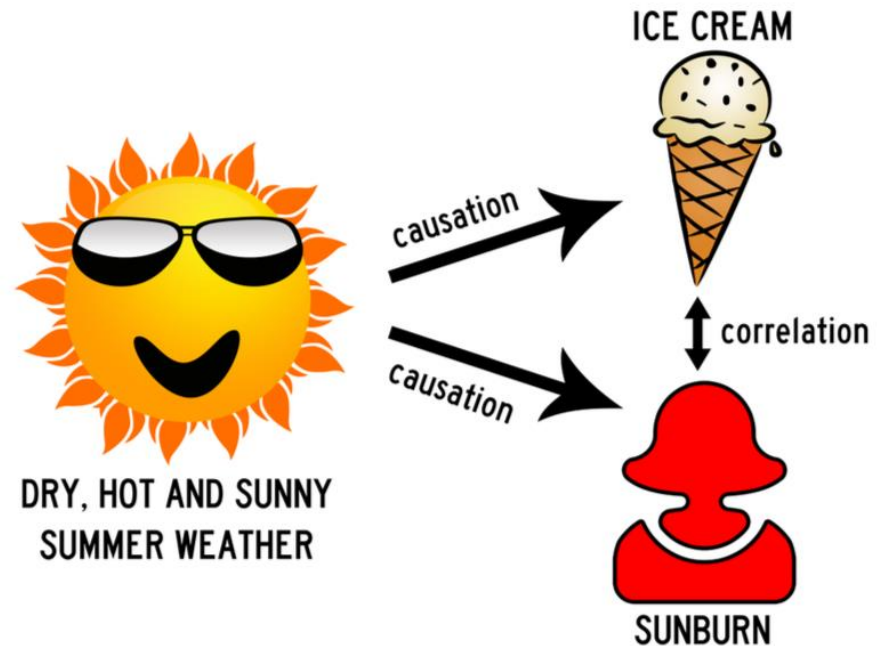


Pitfalls when searching for correlations

Correlation is not the same as causation

The correlation of two variables does not necessarily imply a causal/direct connection. They might both be driven by a “third variable”.

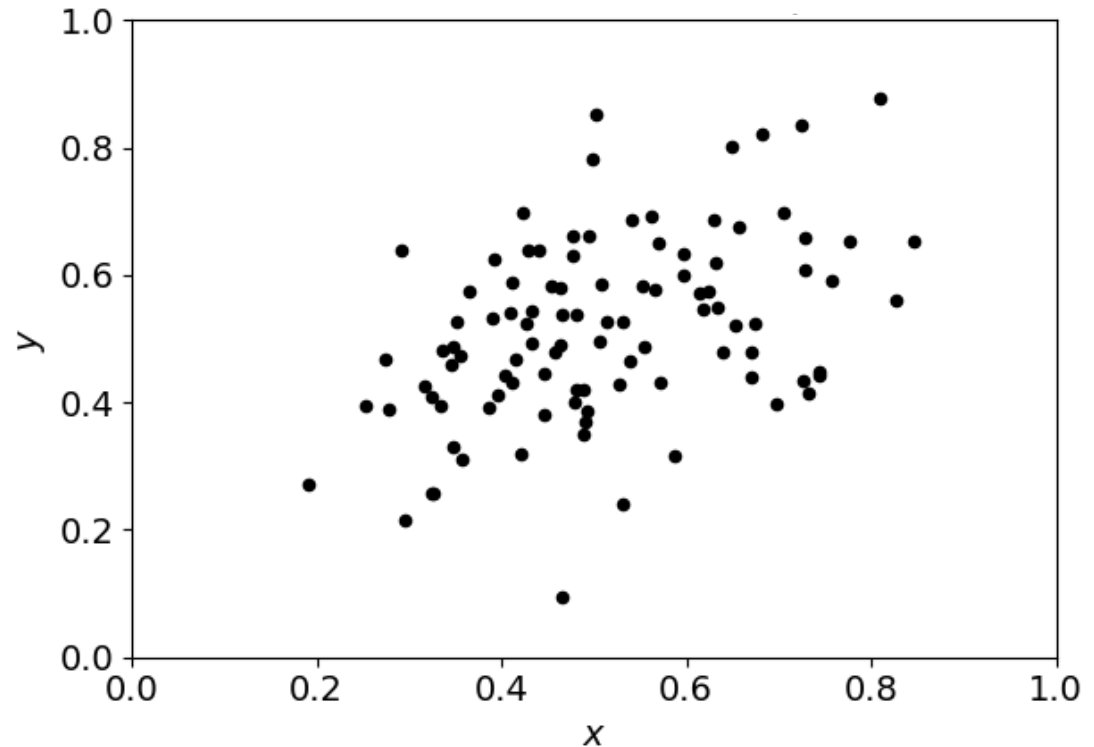
Eating ice cream causes sunburn??



Procrastinate by checking a few more examples at
<https://www.tylervigen.com/spurious-correlations>

Searching for correlations!

*Assuming we
convince ourselves
there are no
selection effects,
systematic outliers,
etc. ... **what next?***



1. What is the **strength** of the correlation?
2. What is the **significance** of the correlation?

Strength of the correlation

We can estimate the correlation coefficient of data samples (x_i, y_i) using the **Pearson product-moment** formula:

$$r = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2 \sum_{i=1}^N (y_i - \bar{y})^2}} = \frac{\sum_{i=1}^N x_i y_i - N \bar{x} \bar{y}}{(N - 1) \sqrt{\text{Var}(x) \text{Var}(y)}}$$

(Or rather, we get python to evaluate this formula!!)

```
: from scipy import stats
r,p = stats.pearsonr(x,y) # evaluate Pearson correlation coefficient r and p-value for non-correlation
print('Correlation coefficient r =',r)
print('Error in r =',np.sqrt((1.-r**2)/float(n-2)))
print('p-value for non-correlation =',p)
```

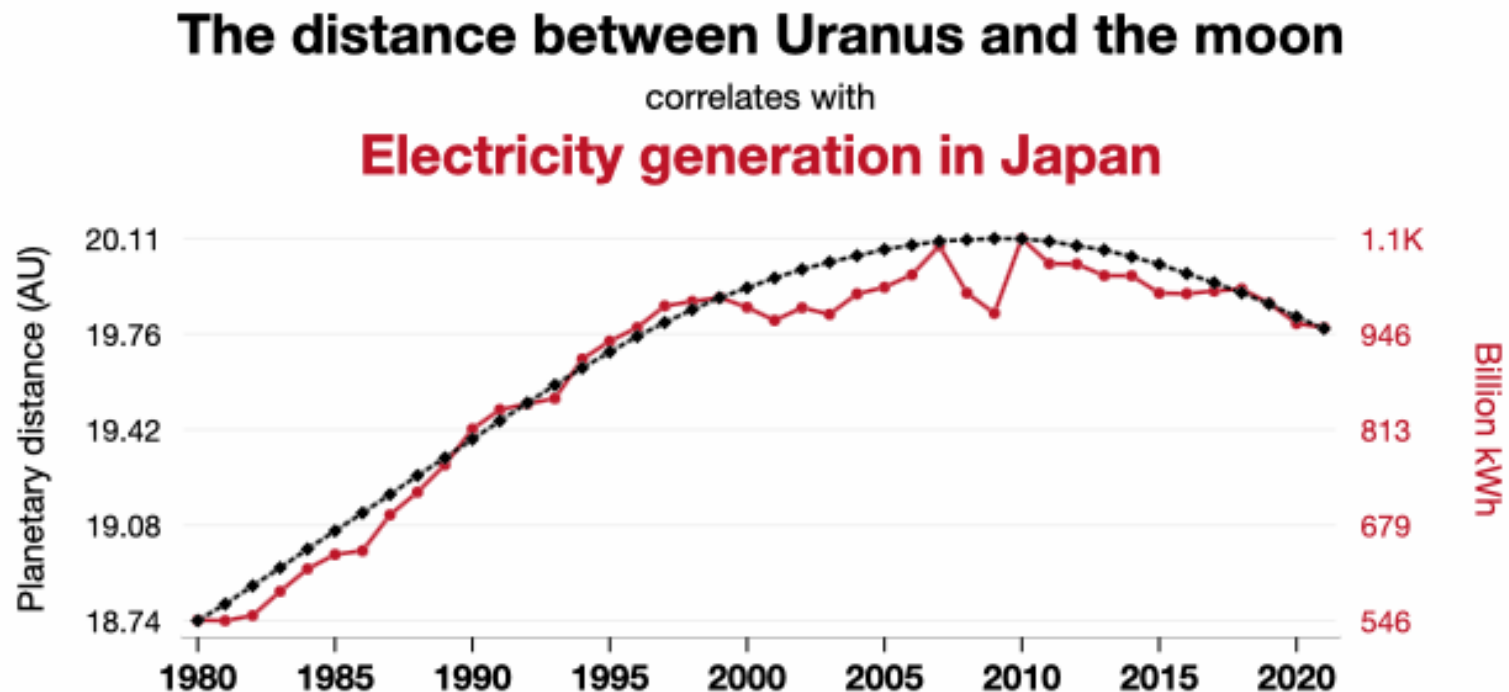
```
Correlation coefficient r = 0.548341521611091
Error in r = 0.08447446031603906
p-value for non-correlation = 3.513624460312244e-09
```

We will get to this in a moment!

The value can range from $-1 \leq r \leq +1$

Significance of correlation

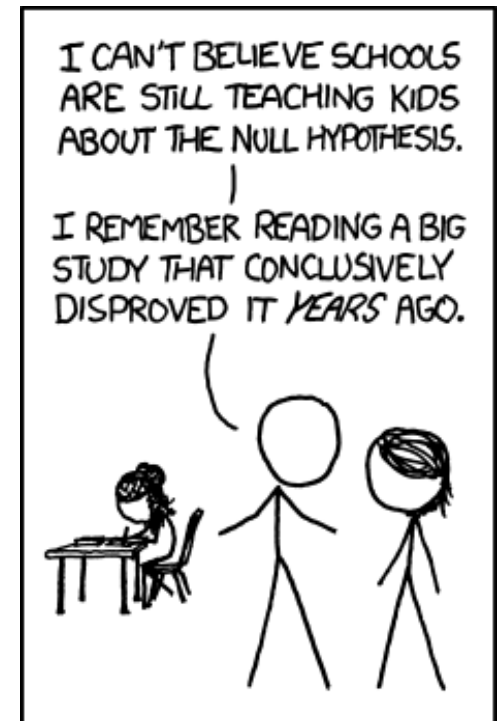
Correlations can arise by
random chance!



Significance of correlation

Correlations can arise by random chance!

- When correlation-testing, it is **not** sufficient to just measure r . We also need to check the **significance of the correlation**
- We can test the significance of the correlation by carrying out a **hypothesis test**



Significance of correlation

How do we apply a hypothesis test to correlations?

- **Null hypothesis:** there is no correlation between the variables

- **Test statistic:**
$$t = r \sqrt{\frac{N - 2}{1 - r^2}}$$

r = correlation coefficient

N = number of data points

(`scipy.stats` is doing all this for us...)

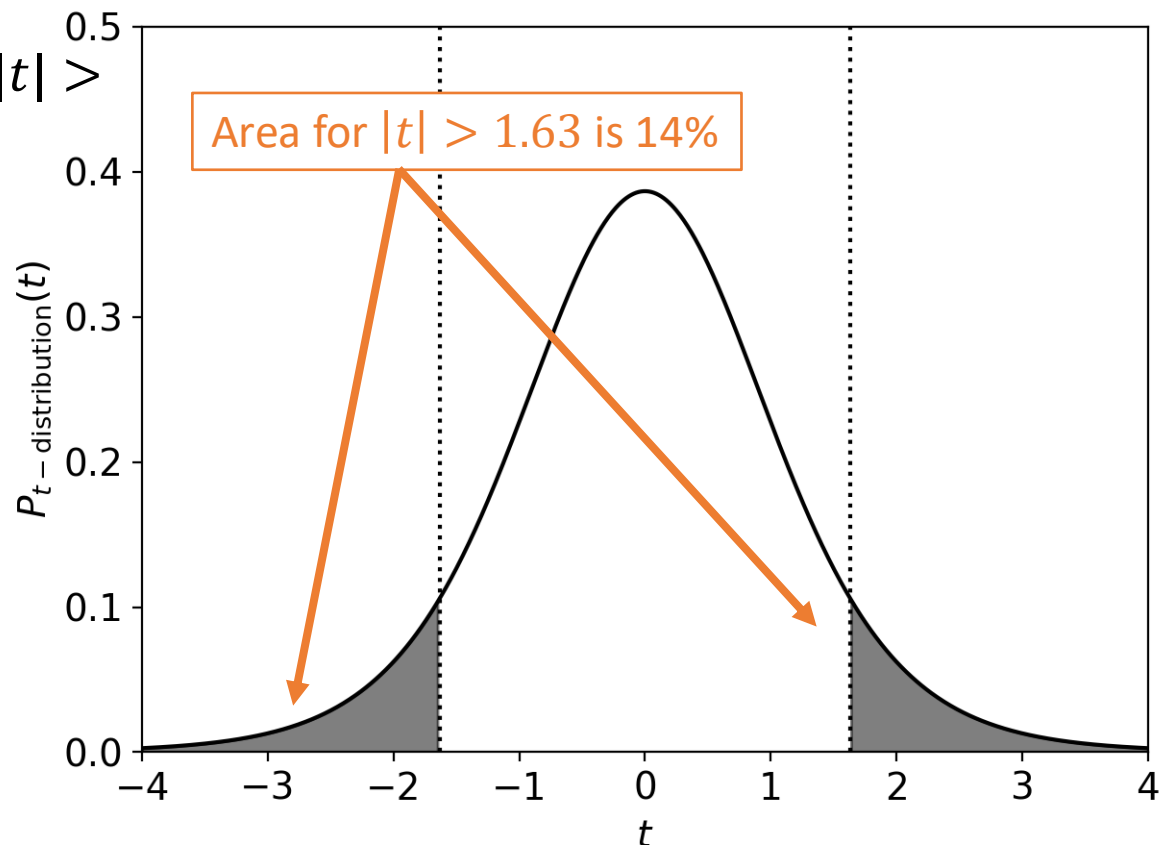
- **Distribution followed by the statistic:** the Student's t probability distribution with number of degrees of freedom $\nu = N - 2$
- **Probability of rejecting the hypothesis:** the area under the tails at higher values of $|t|$ than we have measured

Worked example: correlation test

Example: we measure $r = 0.5$ for $N = 10$ points.
Is this correlation significant?

- We find $t = 1.63$, $\nu = 8$
- The probability of finding $|t| > 1.63$ is **14%**
- This is **not** sufficiently small to reject the hypothesis of no correlation: **this correlation is not significant**
- [we would typically reject with (e.g.) 95, 99% confidence]

If we carried out this experiment many times, this is the distribution that would result!



Activity: Expansion #1

- In this Activity we will check **who discovered the expansion of the Universe!** See Hubble and Lemaitre's distance-velocity datasets. For the two datasets, determine the **Pearson correlation coefficient**, its **error** and **statistical significance**

```
In [5]: # Read in datasets
data = np.loadtxt('../datasets/lemaitre.dat')
dl,vl = data[:,0],data[:,1]
data = np.loadtxt('../datasets/hubble.dat')
dh,vh = data[:,0],data[:,1]
nl,nh = len(dl),len(dh)

# Pearson correlation analysis of Lemaitre dataset
rl,pl = stats.pearsonr(dl,vl)
rlerr = np.sqrt((1.-(rl**2))/float(nl-2))
print('Lemaitre dataset:')
print('r =',rl)
print('error in r =',rlerr)
print('p-value =',pl)

# Pearson correlation analysis of Hubble dataset
rh,ph = stats.pearsonr(dh,vh)
rherr = np.sqrt((1.-(rh**2))/float(nh-2))
print('Hubble dataset:')
print('r =',rh)
print('error in r =',rherr)
print('p-value =',ph)
```

Lemaitre dataset:

- $r = 0.38$
- Error in $r = 0.14$
- $p\text{-value} = 0.013$

Hubble dataset:

- $r = 0.79$
- Error in $r = 0.13$
- $p\text{-value} = 4.5 \times 10^{-6}$

Hypothesis tests and p -values

What is the meaning of this “ p -value” thing??

- The probability of rejecting a hypothesis is often known as a “ p -value”
- It corresponds to the “**significance**” of a result
- *Let's talk about exactly what this value means, since this can be pretty confusing*

<u>P-VALUE</u>	<u>INTERPRETATION</u>
0.001	HIGHLY SIGNIFICANT
0.01	
0.02	
0.03	
0.04	SIGNIFICANT
0.049	
0.050	OH CRAP. REDO CALCULATIONS.
0.051	ON THE EDGE OF SIGNIFICANCE
0.06	
0.07	HIGHLY SUGGESTIVE, SIGNIFICANT AT THE $P < 0.10$ LEVEL
0.08	
0.09	
0.099	HEY, LOOK AT THIS INTERESTING SUBGROUP ANALYSIS
≥ 0.1	

Hypothesis tests and p -values

- Suppose a (no-) correlation significance yields $p = 0.01$
- This means: **there is a 1% chance of obtaining a set of measurements at least this correlated, if the underlying data is uncorrelated** (null hypothesis). It does not mean: 
- “the probability that the points are uncorrelated is 1%” 
- “the probability that the points are correlated is 99%” 
- “if we claim a correlation, there is a 1% chance that we would be mistaken” 
- **Why? Because these are all statements about the model, which we need Bayesian statistics to assess ...**

More correlation tests

But wait, there are more types of correlation test!

Why do we need so many statistical tests??

- Different topics/supervisors have different conventions!
- Different tests may make different assumptions about the data



“If you don't reveal some insights soon, I'm going to be forced to slice, dice, and drill!”

Of course, you shouldn't try every test until you get the “right” result 😊

“Rank” correlation tests

- Hidden assumption behind our previous test: (x, y) data are drawn from a Gaussian. To avoid this assumption ...
- Let (X_i, Y_i) be the rank of (x_i, y_i) in the overall order, such that $1 \leq X_i \leq N$ and $1 \leq Y_i \leq N$
- Compute the **Spearman rank correlation coefficient**

$$r_s = 1 - 6 \frac{\sum_{i=1}^N (X_i - Y_i)^2}{N^3 - N}$$

- Convert the correlation coefficient into a **probability**, using the Student's t distribution as before, with number of degrees of freedom $\nu = N - 2$

Bayesian correlation methods

Yes, we can do a Bayesian version as well ...

- To determine the significance of our correlation, we have been asking, “**what is the probability of measuring a particular value of r if there is no correlation?**”

$$P(r|\rho = 0)$$

- Using Bayesian statistics we can ask the opposite question:

Underlying
parameter (model)

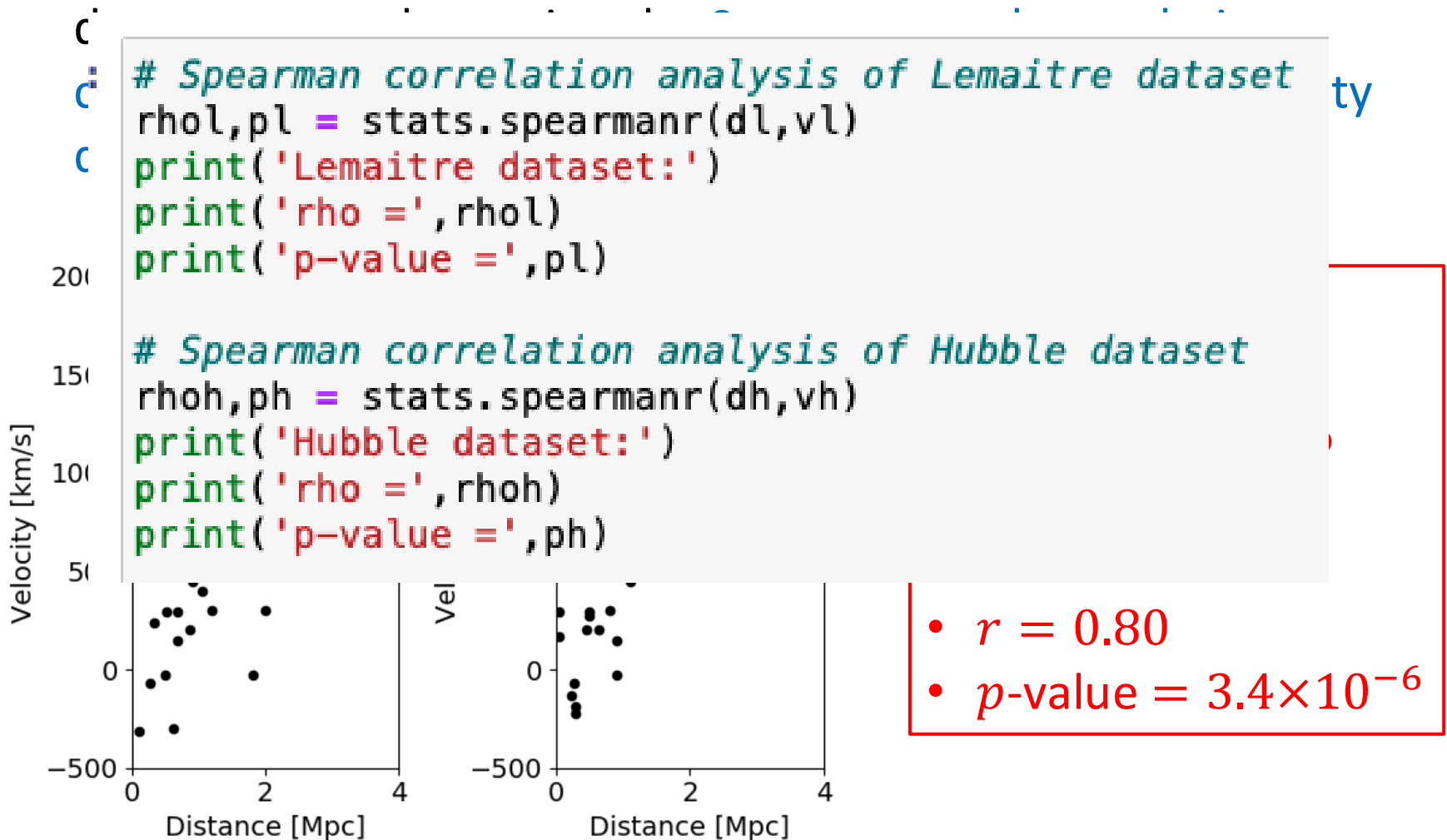
Measured correlation
coefficient (data)

the posterior probability distribution for the
the measured value of r ?”

$$P(\rho|r) \propto \frac{(1 - \rho^2)^{\frac{N-1}{2}}}{(1 - \rho r)^{N-\frac{3}{2}}} \left(1 + \frac{1}{N - \frac{1}{2}} \frac{1 + \rho r}{8} + \dots \right)$$

Activity: Expansion #2

- Returning to Hubble and Lemaitre's distance-velocity



Activity: Expansion #2

- Return
dataset
coeffic
distrib

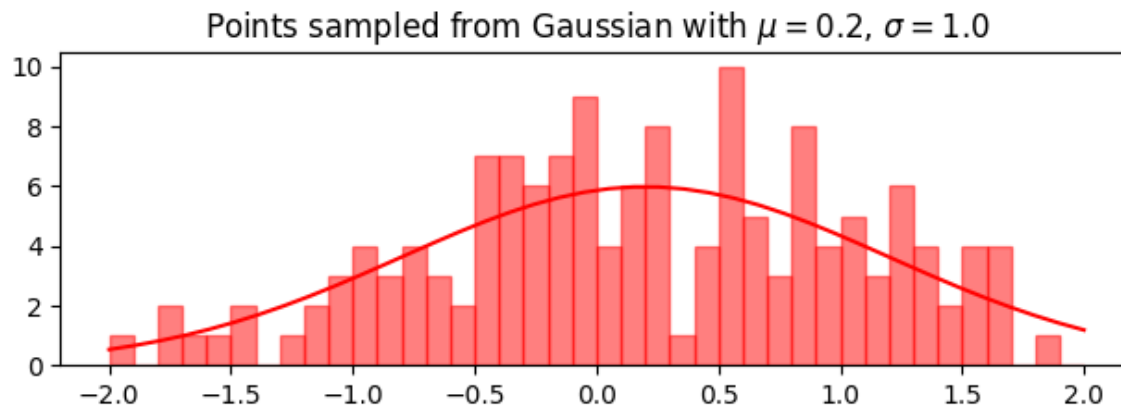
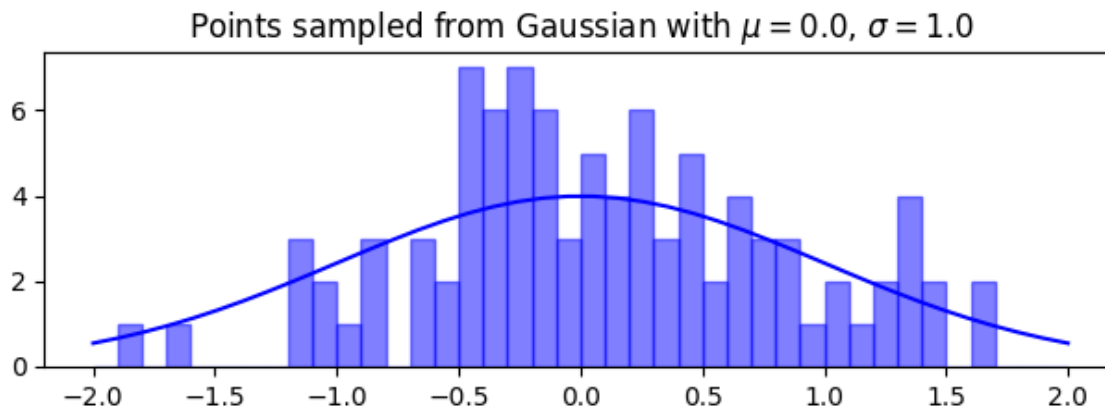
```
# Bayesian posterior probability distributions
xmin,xmax,nx = 0.,1.,101
x = np.linspace(xmin,xmax,nx)
dx = (xmax-xmin)/nx
# Evaluate P(rho|r) for Lemaitre dataset
t1 = (1.-x**2)**(0.5*(nl-1))
t2 = (1.-x*rl)**(nl-1.5)
t3 = (1. + (1./(nl-0.5))*((1.+rl*x)/8.))
probl = (t1/t2)*t3
probl /= np.sum(probl)*dx
# Evaluate P(rho|r) for Hubble dataset
t1 = (1.-x**2)**(0.5*(nh-1))
t2 = (1.-x*rh)**(nh-1.5)
t3 = (1. + (1./(nh-0.5))*((1.+rh*x)/8.))
probh = (t1/t2)*t3
probh /= np.sum(probh)*dx

# Plot the posterior probability distributions
plt.plot(x,probl,color='black',label='Lemaitre')
plt.plot(x,probh,color='red',label='Hubble')
plt.xlabel(r'$\rho$')
plt.ylabel(r'$P_{\rm posterior}(\rho)$')
plt.legend()
```

by
ation
obability

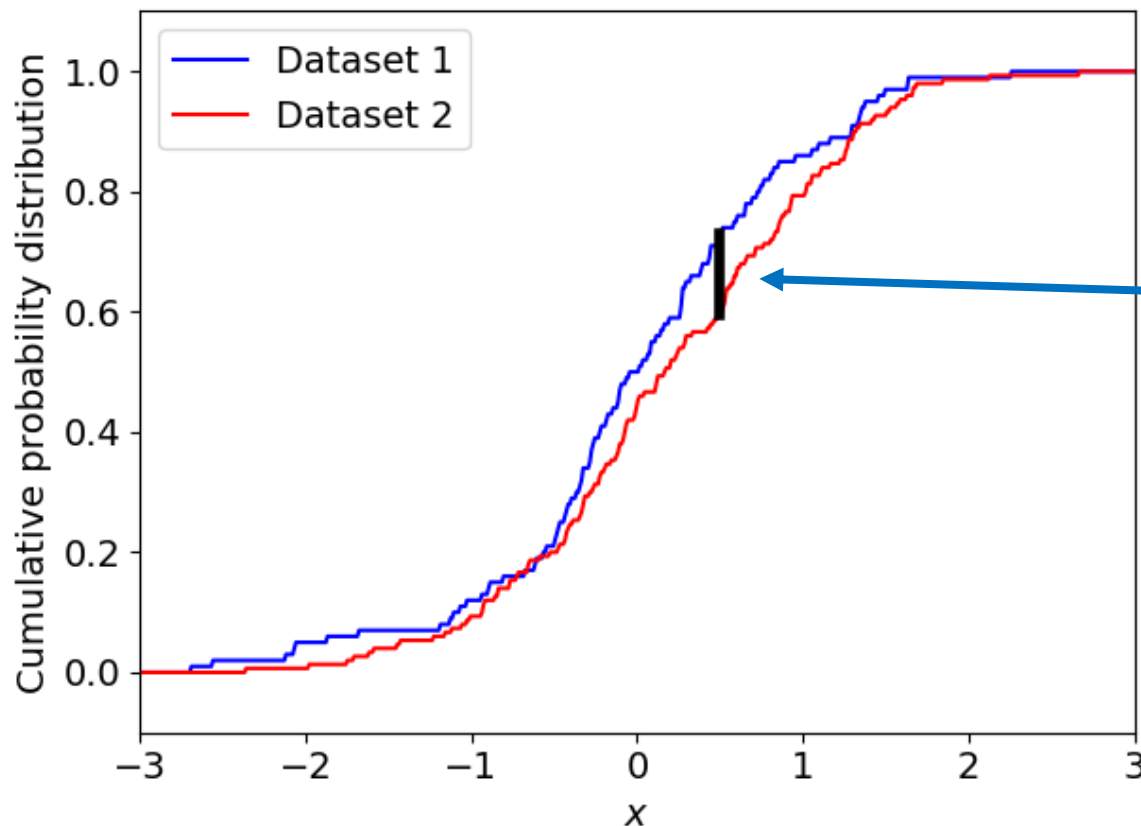
Worked example: K-S test

- To test whether two full distributions are consistent (that is, drawn from the same parent distribution) we can use the **Kolmogorov-Smirnov (K-S) test**



Worked example: K-S test

- Cumulative probability distribution (going from left to right of the histograms):



- The null hypothesis is that these datasets are drawn from the **same parent distribution**
- This is the **maximum deviation**, $d = 0.14$
- The **probability** of rejecting the null hypothesis is $p = 0.196$
- *We confirm the hypothesis!*

Activity: K-S test

- The provided datasets list estimated masses for neutron stars which **are** in double neutron star binaries and are **not** in double neutron star binaries

- Use the t -test to determine whether *there is any significant difference in the means of the*

My answers:

p -value for same means =

- Use
det
ma
con
- ```
datfile1 = '../datasets/ns_masses1.dat'
datfile2 = '../datasets/ns_masses2.dat'
x1 = np.loadtxt(datfile1)
x2 = np.loadtxt(datfile2)
result = stats.ttest_ind(x1,x2)
print('p-value for same means =',result.pvalue)
result = stats.ks_2samp(x1,x2)
print('p-value for same parent distribution =',result.pvalue)
```

# Next week: Model-fitting

